

COMPE 510 – PA 1 Report

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Results

Part 1: Toss a coin

Fprintf result of part 1

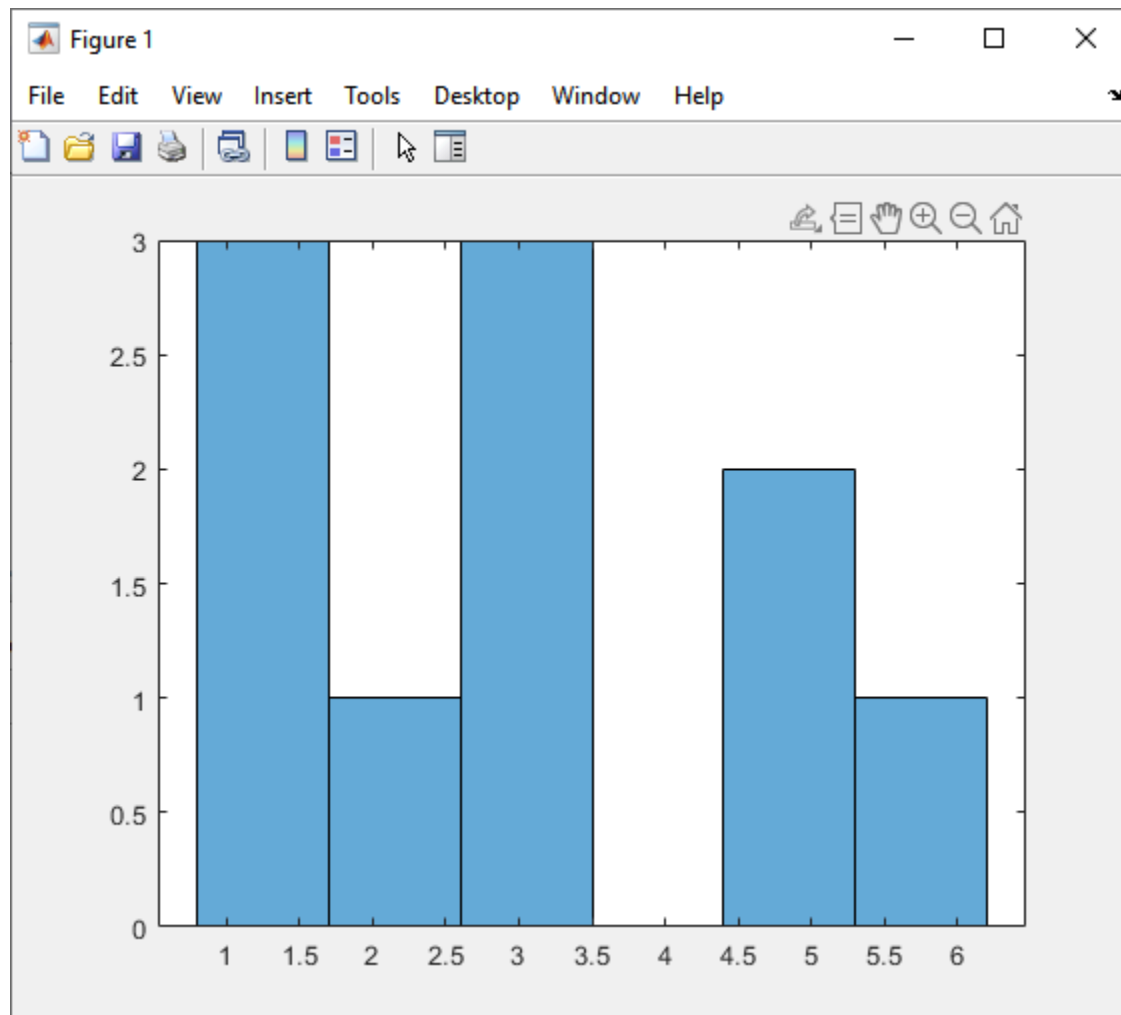
```
Command Window
===== Part 1: Toss a coin =====
Toss a coin ...
The value of the coin:
0
Toss a coin for 10 times ...
The number of heads:
4
Program paused. Press enter to continue.
fx |
```

Part 2 Roll a die:

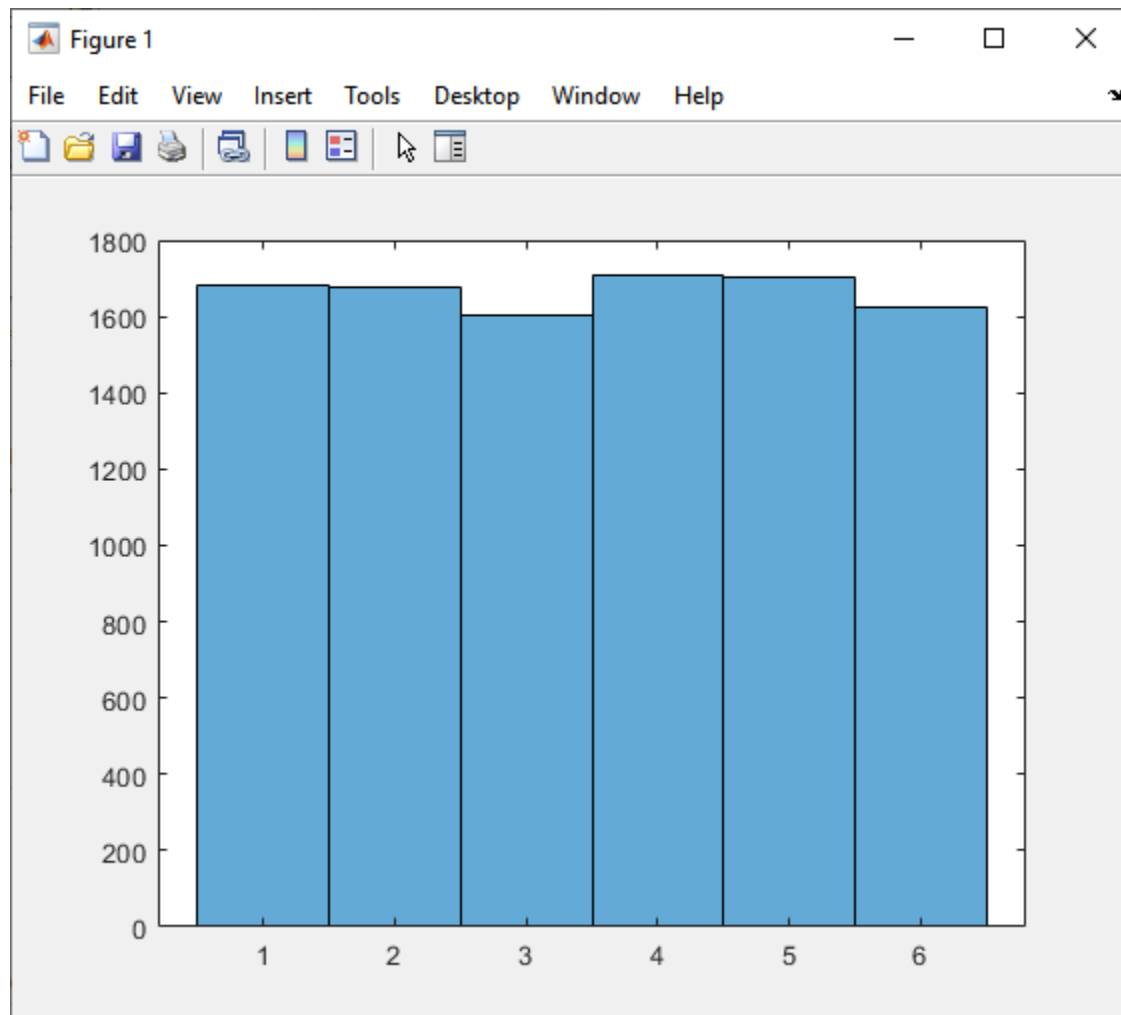
Fprintf result of part 2

```
===== Part 2: Roll a die =====
Roll a die ...
The value of the die:
5
Roll a die 10 times ...
Showing the histogram of the die...
Roll a die 10000 times ...
Showing the empirical cdf of the die...
Program paused. Press enter to continue.
|
```

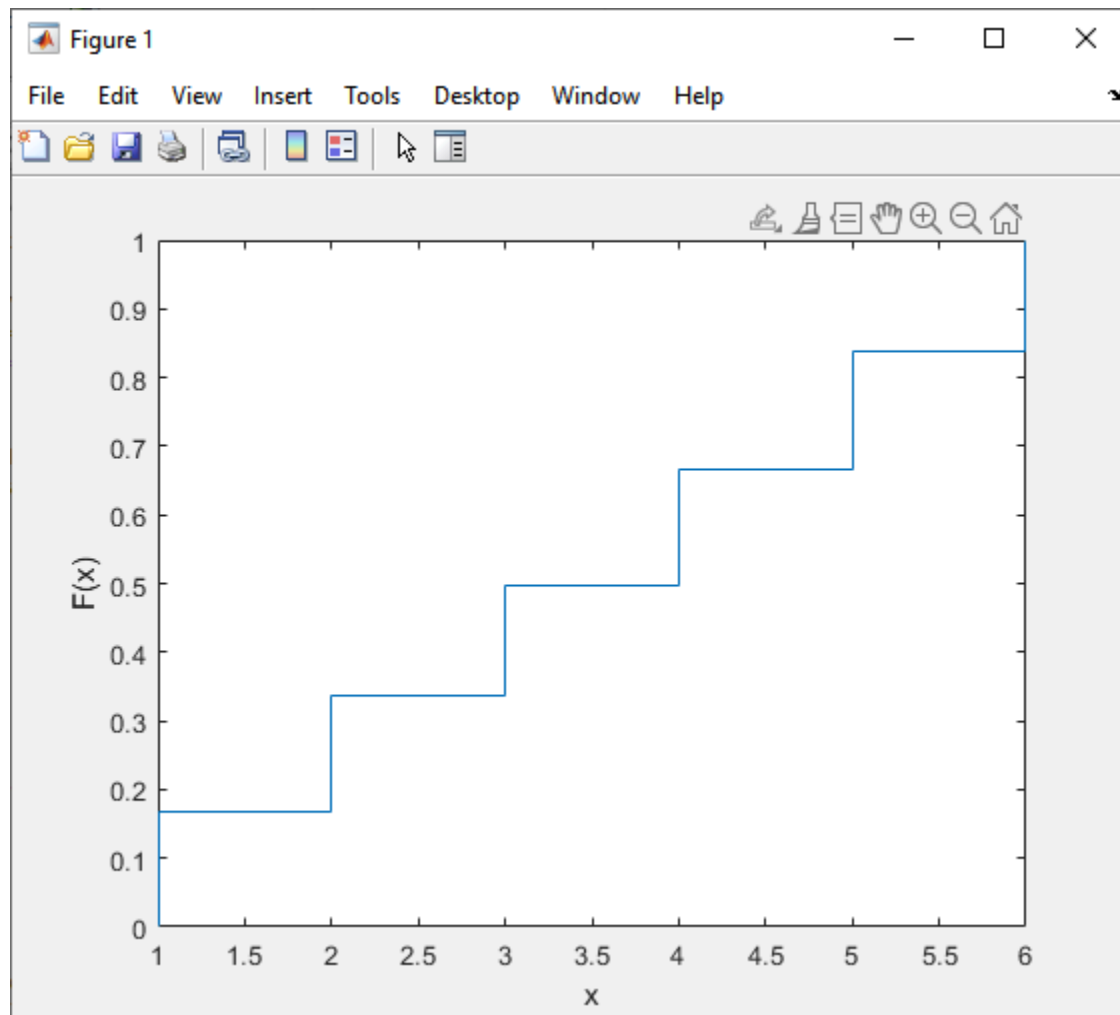
Histogram of rolling a die 10 times



Histogram of rolling a die 10000 times



Empirical cdf of rolling a die 10000 times

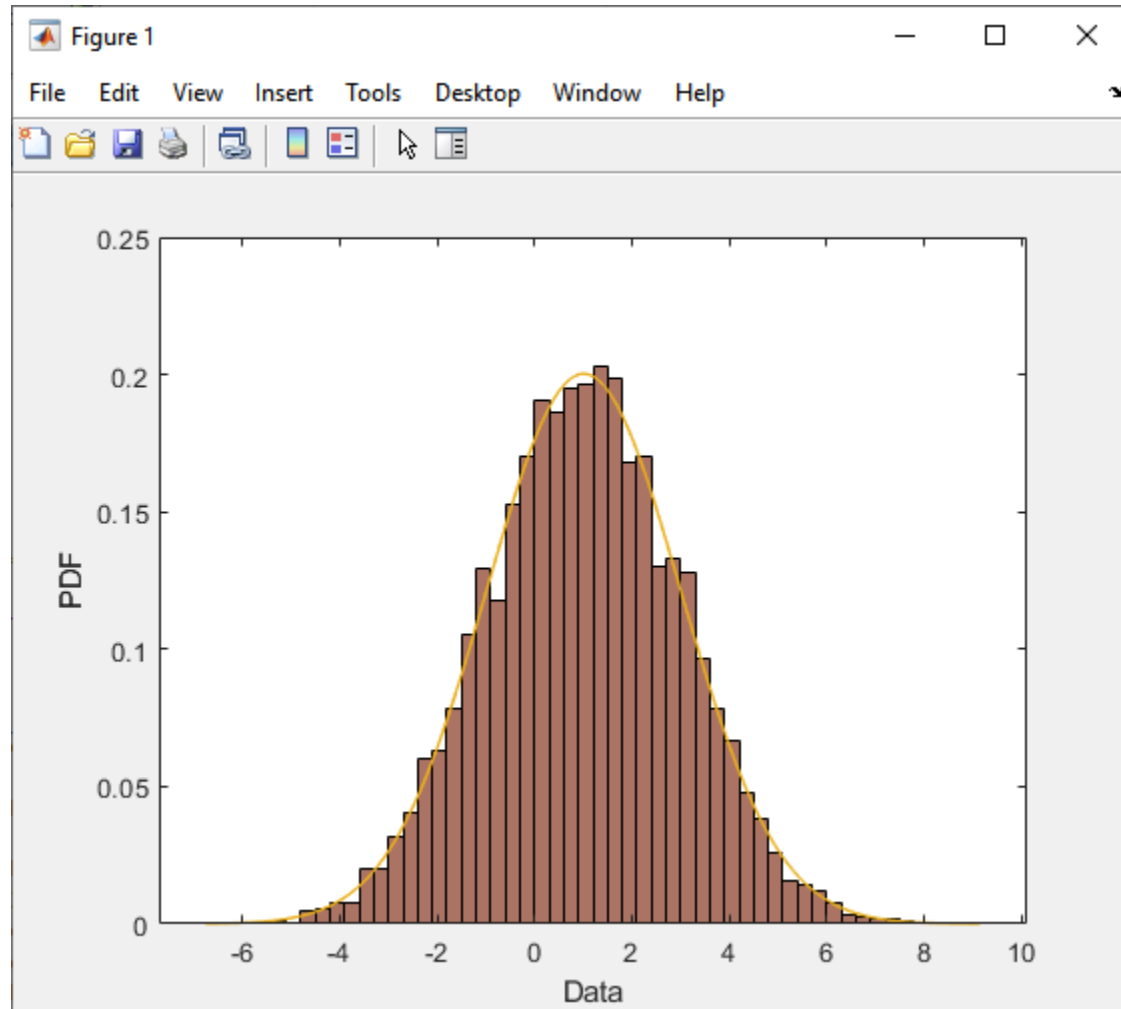


Part 3: Plot a normal distribution

Fprintf result of part 3

```
==== Part 3: Plot a normal distribution ====  
Showing the histogram of the data ...  
Current plot held  
Showing the pdf created by fitting a normal distribution to the data ...  
>>  
>> |
```

Histogram and normal distribution pdf



Questions:

1. In the tossing a coin experiment (part 1), the result of getting 4 heads out of 10 coin flips was what I expected. The most likely outcome according to a binomial distribution, with only two outcomes and equal chance of getting both, would be getting 5 heads; 4 heads is relatively close to 5. If the experiment were repeated multiple times, I would expect the results to change as each experiment is random, but for the results to hover around 5 heads, with especially high amounts of heads or low amounts of heads being less likely to occur.
2. In the rolling a die experiment (part 2), after simulating 10 rolls the results were uneven, with the 6 outcomes getting different unequal amounts. The 10000 rolls in comparison, were significantly more uniformly distributed, only varying by approximately 100 rolls at most. This shows how randomness, through a large sample size, can be more easily analyzed compared to smaller sample sizes, where the underlying distribution is less apparent. Looking at the histograms for part 2, the 10000 rolls has an almost flat distribution, whereas the 10 rolls had an outcome that didn't occur (rolling a 4), and outcomes that occurred more than others (rolling a 1 and 3).
3. I would explain the concept of variance by showing the histogram plot of part 3, pointing out how outcomes occurred mostly around a specific value (in this case the mean of 1), and how as results go further from that value, the less results are present. A pattern to point out would be the width of the plot, and how the wider the plot is the more variance exists, whereas less width represents less variance – values concentrated closer to the mean.

Implementation

Implementing randomness went well, using the Matlab rand feature made it easy to generate random data that could be modified to achieve binary results for the coin flip, or a uniform 6 values for the dice roll.

Making histograms for the data didn't go well during implementation, as the base hist() function in Matlab didn't always display the data correctly. An example being adding tickmarks for 1.5, 2.5, etc. for part 2's 10 dice rolls, despite the fact that those values weren't present in the data which makes the plot less clear (the data is only integers).